

DBMS PRACTICE 4

TOPIC: JOINS AND SET OPERATIONS

STEP 1: CREATE TABLES

```
mysql> create database task4;
Query OK, 1 row affected (0.03 sec)

mysql> use task4;
Database changed
mysql> create table class (
  -> id int,
  -> name varchar(30)
  -> );
Query OK, 0 rows affected (0.06 sec)

mysql> desc class;
+-----+-----+-----+-----+-----+-----+
| Field | Type          | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| id    | int           | YES  |     | NULL    |       |
| name  | varchar(30)   | YES  |     | NULL    |       |
+-----+-----+-----+-----+-----+-----+
2 rows in set (0.01 sec)

mysql>
```

STEP 2: INSERT DATA

```
mysql> insert into class values
  -> (1,'abhi'),
  -> (2,'adam'),
  -> (4,'alex');
Query OK, 3 rows affected (0.03 sec)
Records: 3 Duplicates: 0 Warnings: 0

mysql> select * from class;
+-----+-----+
| id  | name |
+-----+-----+
| 1   | abhi |
| 2   | adam |
| 4   | alex |
+-----+-----+
3 rows in set (0.00 sec)

mysql> |
```

QUESTION 1 - CROSS JOIN

```
mysql> insert into class_info values
  -> (1,'delhi'),
  -> (2,'mumbai'),
  -> (3,'chennai');
Query OK, 3 rows affected (0.03 sec)
Records: 3 Duplicates: 0 Warnings: 0

mysql> select * from class_info;
+-----+-----+
| id  | address |
+-----+-----+
| 1   | delhi   |
| 2   | mumbai  |
| 3   | chennai |
+-----+-----+
3 rows in set (0.00 sec)

mysql> |
```

PART 2: INNER JOIN

```
mysql> create table class (  
  -> id int,  
  -> name varchar(30)  
  -> );  
Query OK, 0 rows affected (0.06 sec)
```

```
mysql> desc class;
```

Field	Type	Null	Key	Default	Extra
id	int	YES		NULL	
name	varchar(30)	YES		NULL	

2 rows in set (0.01 sec)

```
mysql> create table class_info (  
  -> id int,  
  -> address varchar(30)  
  -> );  
Query OK, 0 rows affected (0.06 sec)
```

```
mysql> desc class_info;
```

Field	Type	Null	Key	Default	Extra
id	int	YES		NULL	
address	varchar(30)	YES		NULL	

2 rows in set (0.00 sec)

```
mysql> |
```

```
mysql> insert into class_info values  
  -> (1,'delhi'),  
  -> (2,'mumbai'),  
  -> (3,'chennai');  
Query OK, 3 rows affected (0.03 sec)  
Records: 3 Duplicates: 0 Warnings: 0
```

```
mysql> select * from class_info;
```

id	address
1	delhi
2	mumbai
3	chennai

3 rows in set (0.00 sec)

```
mysql> |
```

```
mysql> insert into class values  
  -> (1,'abhi'),  
  -> (2,'adam'),  
  -> (3,'alex'),  
  -> (4,'anu');  
Query OK, 4 rows affected (0.03 sec)  
Records: 4 Duplicates: 0 Warnings: 0
```

```
mysql> select * from class;
```

id	name
1	abhi
2	adam
3	alex
4	anu

4 rows in set (0.00 sec)

```
mysql>
```

QUESTION 2 - INNER JOIN

```
mysql> select * from class
-> inner join class_info
-> on class.id = class_info.id;
+-----+-----+
| id  | name | id  | address |
+-----+-----+
| 1   | abhi | 1   | delhi   |
| 2   | adam | 2   | mumbai  |
| 3   | alex | 3   | chennai |
+-----+-----+
3 rows in set (0.00 sec)

mysql>
```

QUESTION 3 - INNER JOIN

```
mysql> select class.name,
-> class_info.address
-> from class inner join class_info
-> on class.id = class_info.id;
+-----+-----+
| name | address |
+-----+-----+
| abhi | delhi   |
| adam | mumbai  |
| alex | chennai |
+-----+-----+
3 rows in set (0.00 sec)

mysql>
```

PART 3: NATURAL JOIN

QUESTION 4 - NATURAL JOIN

```
mysql> select*from class
-> natural join class_info;
+-----+-----+
| id  | name | address |
+-----+-----+
| 1   | abhi | delhi   |
| 2   | adam | mumbai  |
| 3   | alex | chennai |
+-----+-----+
3 rows in set (0.00 sec)

mysql>
```

PART 4: LEFT OUTER JOIN

```
mysql> select * from class;
+-----+-----+
| id  | name |
+-----+-----+
| 1   | abhi |
| 2   | adam |
| 3   | alex |
| 4   | anu  |
| 5   | ashish |
+-----+-----+
5 rows in set (0.00 sec)

mysql> select * from class_info;
+-----+-----+
| id  | address |
+-----+-----+
| 1   | delhi   |
| 2   | mumbai  |
| 3   | chennai |
| 7   | noida   |
| 8   | panipat |
+-----+-----+
5 rows in set (0.00 sec)

mysql> |
```

QUESTION 5 - LEFT JOIN

```
mysql> select*from class
-> left outer join class_info
-> on class.id = class_info.id;
+----+-----+----+-----+
| id | name | id | address |
+----+-----+----+-----+
| 1  | abhi | 1  | delhi   |
| 2  | adam | 2  | mumbai  |
| 3  | alex | 3  | chennai |
| 4  | anu  | NULL | NULL    |
| 5  | ashish | NULL | NULL    |
+----+-----+----+-----+
5 rows in set (0.02 sec)

mysql> |
```

QUESTION 6 - LEFT JOIN

```
mysql> select*from class
-> left join class_info
-> on class.id = class_info.id
-> where class_info.id is null;
+----+-----+----+-----+
| id | name | id | address |
+----+-----+----+-----+
| 4  | anu  | NULL | NULL    |
| 5  | ashish | NULL | NULL    |
+----+-----+----+-----+
2 rows in set (0.00 sec)
```

PART 5: RIGHT OUTER JOIN

QUESTION 7 - RIGHT JOIN

```
mysql> select*from class
-> right outer join class_info
-> on class.id = class_info.id;
+----+-----+----+-----+
| id | name | id | address |
+----+-----+----+-----+
| 1  | abhi | 1  | delhi   |
| 2  | adam | 2  | mumbai  |
| 3  | alex | 3  | chennai |
| NULL | NULL | 7  | noida   |
| NULL | NULL | 8  | panipat |
+----+-----+----+-----+
5 rows in set (0.00 sec)

mysql>
```

QUESTION 8 - RIGHT JOIN

```
mysql> select*from class
-> right join class_info
-> on class.id = class_info.id
-> where class.id is null;
+----+-----+----+-----+
| id | name | id | address |
+----+-----+----+-----+
| NULL | NULL | 7  | noida   |
| NULL | NULL | 8  | panipat |
+----+-----+----+-----+
2 rows in set (0.00 sec)

mysql> |
```

PART 6: FULL OUTER JOIN

QUESTION 9 - FULL OUTER JOIN

```
mysql> select*from class
-> left join class_info
-> on class.id = class_info.id;
```

id	name	id	address
1	abhi	1	delhi
2	adam	2	mumbai
3	alex	3	chennai
4	anu	NULL	NULL
5	ashish	NULL	NULL

5 rows in set (0.00 sec)

```
mysql> |
```

QUESTION 10 - FULL OUTER JOIN

```
mysql> select*from class
-> left join class_info
-> on class.id = class_info.id
-> where class_info.id is null
-> union
-> select*from class
-> right join class_info
-> on class.id = class_info.id
-> where class.id is null;
```

id	name	id	address
4	anu	NULL	NULL
5	ashish	NULL	NULL
NULL	NULL	7	noida
NULL	NULL	8	panipat

4 rows in set (0.03 sec)

```
mysql> |
```

PART 7: UNION

```
mysql> select * from first_table;
```

id	ame
1	abhi
2	adam

2 rows in set (0.00 sec)

```
mysql> select * from second_table;
```

id	ame
2	adam
3	chester

2 rows in set (0.02 sec)

```
mysql> |
```

QUESTION 11 – UNION

```
mysql> select * from first_table
-> union
-> select * from second_table;
```

id	ame
1	abhi
2	adam
3	chester

3 rows in set (0.00 sec)

```
mysql> |
```

QUESTION 12 – UNION

```
mysql> select ame from first_table
-> union
-> select ame from second_table;
+-----+
| ame |
+-----+
| abhi |
| adam |
| chester |
+-----+
3 rows in set (0.00 sec)

mysql> |
```

PART 8: UNION ALL

QUESTION 13 - UNION ALL

```
mysql> select * from first_table
-> union all
-> select * from second_table;
+----+-----+
| id | ame |
+----+-----+
| 1 | abhi |
| 2 | adam |
| 2 | adam |
| 3 | chester |
+----+-----+
4 rows in set (0.00 sec)

mysql> |
```

QUESTION 14 - UNION ALL

```
mysql> select count(*)
-> from
-> (
-> select * from first_table
-> union all
-> select * from second_table
-> ) a;
+-----+
| count(*) |
+-----+
| 4 |
+-----+
1 row in set (0.02 sec)

mysql> |
```

PART 9: INTERSECT

QUESTION 15 – INTERSECT

```
mysql> select f.*
-> from first_table f
-> inner join second_table s
-> on f.id = s.id
-> and f.ame = s.ame;
+----+-----+
| id | ame |
+----+-----+
| 2 | adam |
+----+-----+
1 row in set (0.02 sec)

mysql> |
```

QUESTION 16 – INTERSECT

```
mysql> select f.ame
-> from first_table f
-> inner join second_table s
-> on f.ame = s.ame;
+-----+
| ame |
+-----+
| adam |
+-----+
1 row in set (0.00 sec)

mysql> |
```

PART 10: MINUS

QUESTION 17 – MINUS

```
mysql> select f.*
-> from first_table f
-> left join second_table s
-> on f.id = s.id
-> and f.ame = s.ame
-> where s.id is null;
+----+-----+
| id | ame |
+----+-----+
| 1  | abhi |
+----+-----+
1 row in set (0.00 sec)

mysql> |
```

QUESTION 18 – MINUS

```
mysql> select f.ame
-> from first_table f
-> left join second_table s
-> on f.ame = s.ame
-> where s.id is null;
+-----+
| ame |
+-----+
| abhi |
+-----+
1 row in set (0.00 sec)

mysql> |
```

ADVANCED INTERVIEW QUESTIONS

QUESTION 19

```
mysql> select c.id, c.name, ci.address
-> from class c
-> inner join class_info ci
-> on c.id = ci.id;
+----+-----+-----+
| id | name | address |
+----+-----+-----+
| 1  | abhi | delhi   |
| 2  | adam | mumbai  |
| 3  | alex | chennai |
+----+-----+-----+
3 rows in set (0.00 sec)

mysql> |
```

QUESTION 20

```
mysql> select c.id, c.name, case
-> when ci.address is null
-> then 'address missing'
-> else 'address available'
-> end as status
-> from class c
-> left join class_info ci
-> on c.id = ci.id;
```

id	name	status
1	abhi	address available
2	adam	address available
3	alex	address available
4	anu	address missing
5	ashish	address missing

5 rows in set (0.02 sec)

```
mysql> |
```


